

Introductory Econometrics with Recitation — Quiz 3

April 14, 2026

Answer Key

1. **(17 points)** The following table reports estimates from regressions of delivery time, using a sample of 760 food-delivery orders collected from a delivery platform in 2025. The variables are defined as follows:

Variable Name	Description
Time	Total delivery time in minutes.
Distance	Distance from the restaurant to the customer (km).
Traffic	= 1 if the order was placed during a heavy-traffic period, 0 otherwise.
Rain	= 1 if it was raining at the time of delivery, 0 otherwise.
Bike	= 1 if the courier used a bicycle, 0 otherwise.
Downtown	= 1 if the destination is in Downtown, 0 otherwise.
Midtown	= 1 if the destination is in Midtown, 0 otherwise.
Uptown	= 1 if the destination is in Uptown, 0 otherwise.
Suburb	= 1 if the destination is in the Suburb, 0 otherwise.

Regressor	(1)	(2)	(3)
Distance	2.80	2.65	2.50
Traffic	9.40	8.90	8.60
Bike	-4.10	-3.60	-3.40
Rain		6.20	5.90
Downtown			7.10
Midtown			3.80
Uptown			5.20
Intercept	18.60	16.40	14.80
<i>SER</i>	12.50	11.80	10.20
R^2	0.227	0.312	0.487
n	760	760	760

- (a) **(3 points)** Using column (1), do orders placed during heavy-traffic periods take longer, on average, than orders placed outside heavy-traffic periods? By how much?
- (b) **(3 points)** Using column (2), predict the delivery time for an order with a distance of 4 kilometers, placed outside a heavy-traffic period, on a non-rainy day, and delivered by bicycle.
- (c) **(4 points)** Compute and interpret the adjusted R^2 for regression (3).

- (d) (**3 points**) Explain why the regressor *Suburb* is omitted from column (3).
- (e) (**4 points**) Interpret the meaning of *SER* for regression (1) in context. Then, holding the sample and the number of regressors fixed, explain how a decrease in *SSR* affects both *SER* and R^2 .

Suggested Answers for Problem 1:

- (a) Yes. In column (1), the coefficient on *Traffic* is 9.40. This means that orders placed during heavy-traffic periods take, on average, 9.40 minutes longer than orders placed outside heavy-traffic periods.

Grading Criterion

- **(3 pts)**: Correct answer.
- **(0 pts)**: Incorrect answer.

- (b) Using column (2), the regression is

$$\widehat{Time} = 16.40 + 2.65(Distance) + 8.90(Traffic) + 6.20(Rain) - 3.60(Bike).$$

For this order,

$$Distance = 4, \quad Traffic = 0, \quad Rain = 0, \quad Bike = 1.$$

So,

$$\begin{aligned}\widehat{Time} &= 16.40 + 2.65(4) + 8.90(0) + 6.20(0) - 3.60(1) \\ &= 16.40 + 10.60 - 3.60 = 23.40.\end{aligned}$$

Thus, the predicted delivery time is 23.40 minutes.

Grading Criterion

- **(3 pts)**: Correct setup and answer.
- **(1–2 pts)**: Correct method with a minor arithmetic error.
- **(0 pts)**: Incorrect.

- (c) The adjusted R^2 is

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1}(1-R^2),$$

where k is the number of regressors excluding the intercept.

For regression (3), $n = 760$, $k = 7$, and $R^2 = 0.487$. Therefore,

$$\bar{R}^2 = 1 - \frac{760-1}{760-7-1}(1-0.487) = 1 - \frac{759}{752}(0.513) \approx 0.482.$$

So the adjusted R^2 is approximately 0.482.

Interpretation: after adjusting for the number of regressors in the model, regression (3) explains about 48.2% of the variation in delivery time in this sample.

Grading Criterion

- **(2 pts)**: Correct computation $\bar{R}^2$.
- **(2 pts)**: Correct interpretation.

- (d) The regressor *Suburb* is omitted because it is the reference group in the regression. For every order,

$$\text{Downtown} + \text{Midtown} + \text{Uptown} + \text{Suburb} = 1.$$

Therefore, one location dummy must be omitted to avoid perfect multicollinearity.

Grading Criterion

- **(3 pts)**: Correct explanation.
- **(1–2 pts)**: Partially correct.
- **(0 pts)**: Incorrect.

- (e) The standard error of regression is about 12.50 minutes. More precisely,

$$SER = \sqrt{\frac{\sum_{i=1}^n \hat{u}_i^2}{n - k - 1}} = \sqrt{\frac{SSR}{n - k - 1}},$$

where $\hat{u}_i$ is the residual for observation i , and k is the number of regressors excluding the intercept. Also, since

$$R^2 = 1 - \frac{SSR}{TSS},$$

if we hold the sample and the number of regressors fixed, a smaller SSR implies a smaller SER and a larger R^2 .

Grading Criterion

- **(2 pts)**: Correct interpretation of SER .
- **(2 pts)**: correct explanation of its relationship with R^2 .

2. **(13 points)** Suppose that a researcher, using data on 410 walk-in customers and 190 customers with reservations at a popular restaurant, estimates the OLS regression:

$$\begin{array}{rcl} Wait = 42.30 - 13.70 \times Reserve, & R^2 = 0.31, & SER = 10.6 \\ (1.25) & (2.10) & \end{array}$$

where the numbers in parentheses are heteroskedasticity-robust standard errors for the regression coefficients.

The variable *Wait* is the waiting time, measured in minutes, and *Reserve* is a binary variable equal to 1 if the customer had a reservation and 0 if the customer was a walk-in.

- (a) **(4 points)** What is the estimated waiting-time gap between customers with reservations and walk-in customers? What is the average waiting time for each group in the sample?
- (b) **(5 points)** Let $\alpha = 0.05$. Is the estimated waiting-time gap significantly different from 0? Compute the p -value for testing the null hypothesis that having a reservation has no effect on waiting time.
- (c) **(4 points)** Do you think this regression suffers from omitted variable bias? Explain why. Which variables would you add to the regression?

Suggested Answers for Problem 2:

- (a) The estimated waiting-time gap is -13.70 minutes. This means that customers with reservations wait, on average, 13.70 minutes less than walk-in customers.

The average waiting time for walk-in customers is

$$\hat{\beta}_0 = 42.30 \text{ minutes.}$$

The average waiting time for customers with reservations is

$$\hat{\beta}_0 + \hat{\beta}_{\text{Reserve}} = 42.30 - 13.70 = 28.60 \text{ minutes.}$$

Grading Criterion

- **(4 pts)**: Correct answer.
- **(1-3 pts)**: Partially correct.
- **(0 pts)**: Incorrect.

- (b) The hypotheses are:

$$H_0 : \beta_{\text{Reserve}} = 0 \quad \text{and} \quad H_1 : \beta_{\text{Reserve}} \neq 0.$$

The test statistic is

$$z^* = \frac{-13.70 - 0}{2.10} \approx -6.52.$$

The two-sided p-value is

$$p\text{-value} = 2\Phi(-|z^*|) = 2\Phi(-6.52) \approx 0.$$

Therefore, we reject H_0 at the 5% significance level. There is strong statistically significant evidence that having a reservation affects waiting time.

Grading Criterion

- **(5 pts)**: Correct answer.
- **(1-4 pts)**: Partially correct.
- **(0 pts)**: Incorrect.

- (c) Yes, this regression likely suffers from omitted variable bias. For example, waiting time may also depend on party size, whether the customer arrived during peak dinner hours, whether it was a weekend or holiday, and how busy the restaurant was at that time.

Some reasonable variables to add include:

- party size,
- peak-hour indicator,
- holiday indicator,

Grading Criterion

- **(2 pts)**: Reasonable explanation.
- **(2 pts)**: Appropriate omitted variables